Negotiating Design Intelligence: How Reclaimed Material Becomes an Active Agent in AI-Augmented Architectural Design

Keywords

AI-augmented design, human–machine collaboration, machine learning, digital fabrication

Abstract

This paper presents a novel approach to Artificial Intelligence (AI) in architecture, using computation to expand the material and constructive repertoire while fostering new forms of design exploration and human - machine interaction. Featuring a comprehensive digital process chain – including algorithmic design, computational simulation, and robotic fabrication – this approach allows to oversee, model, and control the reuse of building material for circular architectural construction, and, as such, integrates specialized AI-driven models for iterative evaluation, training cycles, and process visualization. In this, the workflow enables (direct) decision-making and intervention at any process step, preserving enhanced digital augmentation, flexibility, and control, and, at the same time, overcoming fully automated process sequences or machinic optimization routines that can fail to address the inherent geometric irregularities and non-standard properties of reclaimed building components.

The study is based on and tested through an experimental demonstrator – the EINZ (Experimental Building for Innovation and Circularity) research project, outlining a new methodological perspective, for which reinforcing steel elements are being used as both construction material and training data for AI models, but also to provide the development and validation framework for the overall AI-augmented design approach. A triangular, space-frame–inspired logic guides the structural system and defines the trained AI models, which generate visual design propositions within given parameters. These outputs are not considered final designs but evolving hypotheses that are evaluated, adapted, and retrained through continuous interaction with fabrication results and material characteristics.

This paper considers a) research parameters and components of AI-augmented design workflows, b) important steps and interfaces of selective design intervention and AI training, c) the implications of automated fabrication and one-to-one validation, and d) the consequences for architectural design, modelling, and knowledge. It explores how designers navigate and negotiate agency across human intention, data-driven inference, and material information, as decision-making becomes distributed among tools, models, and actors. A part of this process is the design of interfaces - ranging from computational datasets and prompts to parametrically controlled fabrication files - through which designers can steer or constrain AI behaviour while retaining authorship and critical judgment.

This methodological convergence transforms architectural modelling from a representational act into a performative and evolutionary process - negotiating between data, matter, and intention. Here, design knowledge is generated through feedback and adaptation. The designer thus operates as mediator, translating between computational inference and material response. Material behaviour becomes an active partner in decision-making, informing both digital model adjustments and AI

retraining. This reciprocal relationship establishes a feedback ecology where material intelligence and human intuition co-evolve, allowing for the emergence of unexpected qualities and solutions that neither the human designer nor the computational model could envision independently.

Ultimately, the integration of all process steps into a systemic, human-centred workflow points toward new epistemic models of designing with (building) materials and the machine. By grounding AI methods in circular, low-tech materials and their inherent affordances, the paper challenges narratives of automation and optimization, instead proposing a situated, responsive mode of architectural intelligence that redefines the dialogue between digital processes, material agency, and design authorship.